



BACHELOR OF INFORMATION TECHNOLOGY (HONS)

**FINAL EXAMINATION
JUNE 2024**

Course : EC3123 (Operating Systems Principles) Time: 2.00 pm – 5.00 pm (3 hours)
Lecturer : Mohd Shahdi Bin Ahmad Date: 27 September 2024

Instructions:

Answer **ALL** questions in the Answer Booklet provided.

The maximum number of marks is 100.

This question paper consists of 2 printed pages.
(excluding front cover)

DO NOT OPEN THIS BOOKLET UNTIL YOU ARE TOLD TO DO SO

- 1 a. What is the primary function of an operating system? (2 MARKS)
- b. Create a diagram that illustrates the role of the four essential managers of an Operating System. (10 MARKS)
- c. Describe how the essential managers in question Q1(b) interact with each other. (8 MARKS)
- d. Identify FIVE (5) types of Operating Systems. (5 MARKS)
- 2 a. State transitions refer to the changes in the condition or mode of a system or process. Draw the state transition diagram. (11 MARKS)
- b. Explain the transitions between the states in question Q2(a). (10 MARKS)
- c. Process creation is a fundamental operation in operating systems, involving generating a new process from an existing one. Draw the process creation with the correct entities. (4 MARKS)
- 3 a. A process can be swapped momentarily from memory to a backup store, and then it can be returned to memory to continue executing. Draw the swapping procedure using the statement as a guide. (8 MARKS)
- b. What is the purpose of paging in memory management? (2 MARKS)
- c. How does paging divide memory? (2 MARKS)
- d. Draw the paging model of logical and physical memory. (8 MARKS)
- e. Identify FIVE (5) logical units in segmentation. (5 MARKS)

- 4 a. Draw the layers of the I/O system in user-space I/O software. (10 MARKS)
- b Explain the function of each layer in question Q4(a). (10 MARKS)
- c. Identify FIVE (5) file attributes in a file system. (5 MARKS)

Device Driver

Interrupt Handler

-END OF QUESTION PAPER-